

DiaTool-DPO

Enhancing Tool-Augmented LLM Dialogue via Direct Preference Optimization

A novel DPO-based method modeling TA-LLM interactions as an MDP with 5 dialogue states and 3 query types

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

TA-LLMs trained only with SFT struggle with incomplete queries and out-of-scope requests, leading to hallucinated tool calls.

1

Missing Follow-Up Questions

Model fails to ask clarifying questions when required parameters are missing from user queries

2

Hallucinated Tool Invocations

Model makes tool calls with fabricated or incorrect argument values instead of requesting proper slot-filling

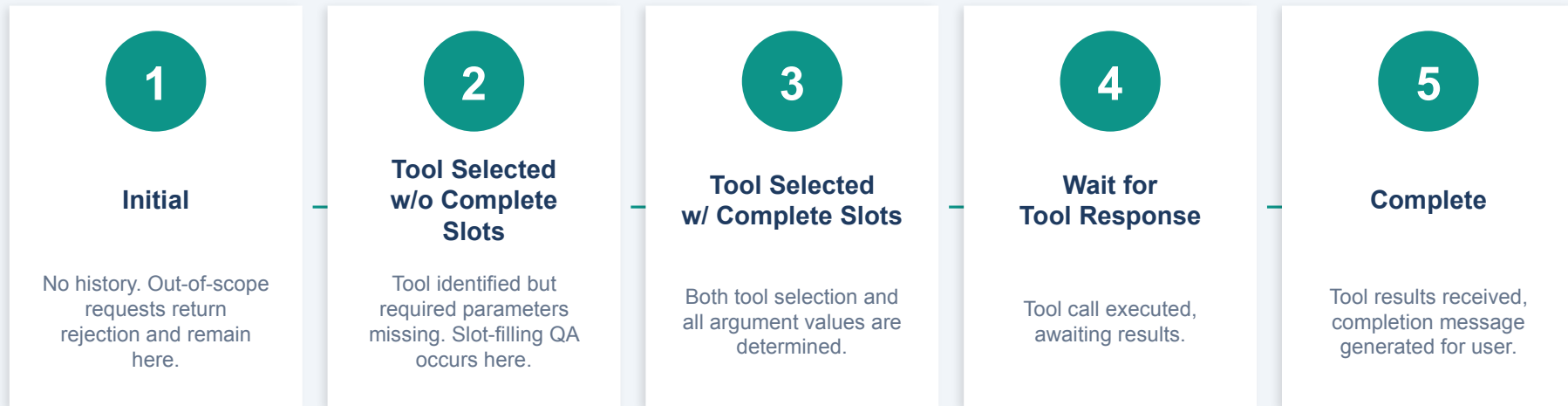
3

Failed Tool Call Rejection

Model attempts to use unavailable tools instead of properly rejecting out-of-scope requests

Solution: Use Direct Preference Optimization to automatically construct paired trajectories of correct vs. incorrect dialogue flows

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process



Key Insight

The 5 internal states are unobserved — the model must learn to navigate them based on user queries and tool responses. This MDP formulation enables systematic construction of preference pairs for DPO training.

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Condition	Example
Type 1	1 → 3 → 4 → 5	Sufficient info provided	"Book a flight from NYC to LA on April 15"
Type 2	1 → (2×N) → 3 → 4 → 5	Incomplete info, needs slot-filling	"Book a flight" (missing destination, date)
Type 3	1 → 1	Out-of-scope request	"Call my mom" (no tool available)

Note: State 2 repeats N times until all required slots are filled. Type 2 trajectories demonstrate the model's ability to engage in multi-turn slot-filling conversations.

Query	Chosen Traj.	Rejected Traj.	Learning Lesson	Count	Difficulty	Without
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy	Automatically by comparing
	1→3→4→5	1→(2*N)→3→4→5	Prevent redundant slot-filling	562	Hard	
	1→3→4→5	1→1	Tool call accept	2,090	Easy	
	Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	

13,138

Total Pairs

3

Query Types

0

Human Labels Required

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

SFT + DiaTool-DPO consistently achieves highest Macro Average across all model sizes

LLaMA-3-8B: Dramatic Gains with DiaTool-DPO



Slot Accuracy

0.639 → 0.917

+43.5%

Relevance

0.826 → 0.913

+10.5%

These improvements bring LLaMA-3-8B to near GPT-4o performance levels (94.8% info gathering, 91% tool call rejection) while running on consumer hardware.

Key Takeaways

RL-Based Dialogue Control for Production TA-LLMs

1

MDP formulation enables systematic trajectory pairing — defining 5 internal states and 3 query types allows automatic generation of chosen/rejected trajectory pairs without human annotation

2

DiaTool-DPO consistently improves Macro Average — across all three model sizes (3.1B, 8B), SFT + DiaTool-DPO achieves the best balanced performance

3

Slot-filling and tool rejection see the largest gains — LLaMA-3-8B improves Slot accuracy by +43.5% and Relevance by +10.5% over SFT-only

4

DPO alone is insufficient — DiaTool-DPO-only performs poorly, confirming that DPO complements rather than replaces supervised training

5

Language-agnostic approach — while primary experiments were in Korean, the method generalizes to English, opening broad applicability for production TA-LLMs

DiaTool-DPO opens a path to GPT-4o-level conversational tool use on smaller, deployable open-source models